

PROJECT JANUS

Adversarial Shadow-World Architecture (ASWA) - Technical Blueprint

1. Executive Summary

Project Janus introduces ASWA, a defensive framework that discovers risk in an isolated Shadow-World twin and returns only validated defenses through the Janus Gate. Production (Light World) is never attacked directly.

2. Core Components

RME: mirrors and sanitizes production telemetry into a twin.

ASA: probes policy only inside the Shadow-World using a closed action catalog.

Janus Gate: one-way validation and install of defensive patches only.

OpenAI LLM (optional): adversary plan proposals, Learn rationales, operator briefing.

3. Operational Loop

1) Duplicate 2) Attack 3) Learn 4) Deploy

Graded baseline uses Q-learning ASA with no API key.

Optional OpenAI LLM roles activate when OPENAI_API_KEY is present in .env.

4. Containment

Secrets are redacted before twin hydration.

ASA and OpenAI LLM adversary stay inside catalog actions and the twin.

Gate blocks payload export and installs defense rules only.